

Nested-quotient temporal histories and homological flat bands in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

Final edition, with machine-checked provenance

Alexander Smith

Independent Researcher

29 August 2026

Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice at strong coupling. After selecting the complete gauge-invariant charge-odd one-plaquette eigenspace, fixed-order contributions are organised as ordered magnetic histories on the Haar–Gram quotient of reachable Wilson networks. The complete momentum dependence through second order is the oriented edge–face Gram operator $t_N u^2 B(k) B(k)^\dagger$ with $B = \partial_2^\dagger$ and

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0, \quad N \geq 3.$$

The projected spectrum is $\{0, t_N u^2 q, t_N u^2 q\}$ up to a common scalar, with $q = 4 \sum_j \sin^2(k_j/2)$; the flat carrier is $\ker \partial_2$, of exact dimension $L^3 + 2$.

Relative to the 28 August edition this one adds four things, each carrying its own machine check. The channel decomposition of t_N , previously a representation-theoretic ledger derived on a single cube, is recovered on a genuine face-sharing two-cube $SU(3)$ space of 1,590,462 states, and the two decompositions agree *channel by channel* rather than only in the sum (Theorem 12). Separately, and by arithmetic on the weight ledger alone, a cutoff keeping only the singlet and $\Lambda^2 F$ routes projects the hopping to $-1/12$ — opposite in sign to t_3 — and omits exactly 14/153, the two summing to t_3 (Corollary 14); the $B = 4$ truncation of the two-cube space is shown to retain the same routes, conditional on the same delivery. The charge-even sector acquires a closed form: with $\hat{a}_m = 2 + 2 \cos k_m$ and $p = \sum_m \hat{a}_m = 12 - q$, its Bloch spectrum is the cubic $\mu(\mu - p)^2 = 4\hat{a}_1 \hat{a}_2 \hat{a}_3$, so one zone function runs both sectors (Theorem 17), and its Γ -point splitting pins the charge-even hopping $t_{r,+}$ — a separation the charge-odd rest frame has no analogue for, since its Γ spectrum has only one level (Proposition 20). Finally the open fourth-order off-axis coefficient is given a geogra-

phy rather than a verdict: an exact obstruction certificate, an explicit second witness, and the measurement that no Γ -anchoring error can move it at all (Section 9).

$SU(3)$ third- and fourth-order statements remain conditional on supplied exact-arithmetic ledgers whose raw payloads are not reproduced here, and are labelled as such throughout. Every displayed result carries the name of the machine check that establishes it. All statements concern a formal finite-order projected sector: this is not a continuum glueball or mass-gap theorem.

1 Introduction

Hamiltonian strong-coupling expansions organise lattice gauge dynamics around the electric vacuum and its flux excitations [1, 2]. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and the resulting interference is the central object here.

Flat bands generated by incidence or Gram matrices have a long history: local topology, line-graph constructions, compact localised states, and the need for noncontractible states on periodic systems are all established, and closely related closed-surface structures occur in two-form spin liquids. A recent classification of exactly flat bands protected by local graph topology [14] reaches extensive face-graph degeneracy from an *unoriented* face–vertex incidence, where rank–nullity alone gives $|F| - |V|$ flat bands. That is a different mechanism from the one here, and the difference is measurable: on the signed face–edge operator of Section 3 the corresponding index is exactly 0 and bounds nothing, while both kernels equal $L^3 + 2$ — homology, not counting. It is recorded as a distinction, not an identification.

We do not claim that incidence flatness, cellular compact localised states, or homological counting are new in isolation.

check: the discrete index theorem gives 0 on the signed face-edge operator, bounding nothing

The content is dynamical. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, exact strong-coupling matrix elements generate an oriented edge-face Gram operator in the charge-odd one-plaquette sector. Representation theory fixes its coefficient and the cubical chain complex fixes its kernel and spectrum. The separation is exact,

$$\begin{aligned} \text{colour dynamics} &\longrightarrow t_N, \\ \text{cellular geometry} &\longrightarrow BB^\dagger. \end{aligned}$$

What is proved, and what is conditional

Six statements are self-contained here.

1. For every $N \geq 3$ the shared-link channel weights are d_ρ/N^2 , derived from the order-two Weingarten values (Theorem 6). This was previously an isotropy *hypothesis*, and everything in Section 4 rests on it.
2. For every $N \geq 3$ the complete momentum dependence at order u^2 is $t_N u^2 BB^\dagger$ (Theorem 9).
3. The Bloch spectrum of $BB^\dagger$ is $\{0, q, q\}$ (Theorem 1), and on the periodic L^3 complex the zero-mode count is $L^3 + 2$ (Theorem 2). The two are the same statement, by a sum rule over the momentum grid (Theorem 4).
4. The truncation that keeps only the singlet and $\Lambda^2 F$ routes projects the hopping to $w_{\Lambda^2 F} - w_1 = -1/12$, and what it omits is $w_{\text{Sym}^2 F} - w_{\text{Adj}} = 14/153$; the two sum to t_3 (Corollary 14). This is arithmetic on (11) and needs no input beyond Theorem 6.
5. The charge-even Bloch spectrum is the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$ with $p + q = 12$ (Theorem 17); its range is exactly $[-4, 12]$, the top attained only at Γ and the floor on the three planes $k_j = \pi$, triple only at R (Proposition 18); and its Γ -point splitting is $12t_{r,+}$ exactly, at both computed orders (Proposition 20).
6. The retained Γ /axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 25), and an explicit second witness exhibits the non-identifiability by construction (Proposition 27).

Two families of statements are *conditional* on supplied exact-arithmetic derivation ledgers. Their raw class contractions, symbolic history certificates and microscopic rank-sweep artifacts are not included here,

and repeated assertions in synthesis documents are not treated as independent reproductions.

7. A supplied two-cube delivery reports six link-irrep channel coefficients at $N = 3$, as rational reconstructions of finite-precision contractions. Conditional on those six numbers, they are the four fusion-channel weights individually — channel by channel and not merely in the sum (Reported result 12) — and the $B = 4$ truncation of that space is the same truncation as the published link cut-off of item 4. The 1,590,462-state build was not reproduced here.
8. For $SU(3)$, supplied ledgers report that all candidate third-order lifter classes vanish and report the scalar and shared-link coefficients of Section 7. The census is three lifter classes evaluated over 32 five-trace numerator patterns each, 96 contractions in total, and nothing here checks any of them. If it is exhaustive and the linked ledger complete, the effective Hamiltonian stays in the incidence algebra through order u^3 .
9. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2 - 1)^3]$ and its equality with the complete axial coefficient for $N = 3, \dots, 12$. Conditional on those calculations, the $SU(3)$ carrier acquires nonzero axial dispersion at order u^4 . The complete off-axis operator remains open — and by Theorem 25 it cannot be closed by any Γ or axial datum.

These statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but the external spectral problem is not the full gauge-theory Hilbert space. At $k = 0$ the three incidence branches touch and the finite-volume separation collapses as L^{-2} . Nothing below proves convergence of the strong-coupling series, persistence under unrestricted higher-order operators, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling the Kogut–Susskind Hamiltonian is

$$H_\beta = \frac{1}{2} \sum_\ell C_2(\ell) + \beta_N \sum_p \left(1 - \frac{1}{N} \text{Re Tr } U_p\right). \quad (1)$$

Dropping the additive constant,

$$H = H_E - uV, \quad V = \sum_p (\chi_p + \bar{\chi}_p), \quad u = \frac{\beta_N}{2N}, \quad (2)$$

with $\chi_p = \text{Tr } U_p$. For $SU(3)$, $u = \beta_3/6$. The definition $u = \beta_N/(2N)$, not the $SU(3)$ -specific equality, is the all-rank convention; the archived line $Y = 2\beta/3 = 4u$ is a definition-label erratum and the printed coefficients are already in u . Reading it as a coordinate would demand dividing the order- r coefficient by 4^r , which misses the printed tables by exactly 16 at u^2 and 64 at u^3 .

check: the printed towers are canonical-u: 4*Delta(3u/2) reproduces them verbatim

check: the 4**r rescaling breaks the bridge: order 2 off by 16, order 3 by 64

All quoted excitation energies use the linked, vacuum-subtracted operator $H_{\text{exc}}(u) = H(u) - E_{\text{vac}}(u)I$. This changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

The electric vacuum $|0\rangle$ carries the trivial representation on every link. For $N \geq 3$ the normalised one-plaquette states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}} |0\rangle, \quad (3)$$

of unit norm by character orthogonality. Each of the four boundary links carries the fundamental or antifundamental representation, so

$$E_{F,N} = 4\left(\frac{1}{2}C_F\right) = 2C_F = \frac{N^2 - 1}{N}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (4)$$

check: each channel gap is C_F + C_R/2, and the weights sum to one

For $SU(2)$ complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$. The same fact appears again in Section 4: $\Lambda^2 F$ is the singlet exactly at $N = 2$, which is where t_N vanishes. The charge-even hopping does *not* vanish there, which locates the $N^2 - 4$ zero in the channel difference rather than in the incidence.

check: at $N = 2$ the C-odd hopping vanishes and the C-even one does not

We work inside the charge-odd fundamental Wilson-loop cyclic sector with trivial electric centre flux around each torus cycle. Let P_- be the full $E_{F,N}$ spectral projector there. For $L \geq 3$ its range is the charge-odd one-plaquette span, since a reduced contractible fundamental loop at that energy has four distinct links and is an elementary plaquette. This sector qualification also excludes the accidentally degenerate straight wrapping loop at $L = 4$. With $Q = 1_- - P_-$ the reduced resolvent crosses no omitted $E_{F,N}$ eigendirection, and degenerate perturbation theory gives

$$H_{\text{eff},-} = P_- H_E P_- + uH_1 + u^2 H_2 + u^3 H_3 + \cdots \quad (5)$$

with every H_r linked, vacuum-subtracted, and carrying the appropriate Q -space resolvents.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$ with $\partial_2 \partial_3 = 0$. One-plaquette amplitudes live in C_2 and their signed leakage into links is $\partial_2 c$: a cellular closed-surface condition, not the microscopic vertex Gauss law, which the Wilson-loop basis already satisfies through $\partial_1 \partial_2 = 0$.

Writing $d_j = e^{ik_j} - 1$ and $q(k) = \sum_j |d_j|^2 = 4 \sum_j \sin^2(k_j/2)$, the face-to-link Bloch incidence matrix is

$$B(k) = \partial_2(k)^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \quad (6)$$

Theorem 1 (Incidence factorisation). *With $w(k) = (\bar{d}_3, -\bar{d}_2, \bar{d}_1)^\top$,*

$$BB^\dagger = qI - ww^\dagger, \quad B^\dagger w = 0, \quad \|w\|^2 = q,$$

and therefore $\text{spec}(BB^\dagger) = \{0, q, q\}$. With $S = BB^\dagger - 4I$ the signed shared-link adjacency, $\text{spec } S(k) = \{-4, q - 4, q - 4\}$.

Proof. Multiplying (6) by its adjoint gives the first identity entry by entry. Since $\|w\|^2 = q$, the rank-one operator $qI - ww^\dagger$ annihilates w and acts as qI on its orthogonal complement. Subtracting $4I$ gives the adjacency spectrum. $\square$

check: B B^dagger = q I - d conj(d)^T for the curl incidence

For $k \neq 0$ the carrier is one-dimensional and represented by $w(k)$. At Γ , $B = 0$ and all three branches meet; the normalised Bloch eigenvector has no direction-independent limit there, which is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localised basis.

Theorem 2 (Finite-torus carrier). *For the periodic cubic cellulation of T^3 , $\ker \partial_2 = \text{im } \partial_3 \oplus H_2$ with $\dim \text{im } \partial_3 = L^3 - 1$ and $\dim H_2 = 3$, so $\dim \ker \partial_2 = L^3 + 2$.*

Proof. There are L^3 three-cells and no four-cells, and $b_3(T^3) = 1$, so $\text{rank } \partial_3 = L^3 - 1$ by rank-nullity. Discrete Hodge decomposition gives $\ker \partial_2 = B_2 \oplus H_2$ with $\dim H_2 = b_2(T^3) = 3$. $\square$

The count is verified over $\mathbb{Z}$ rather than re-derived from the formula: the complex is built from the boundary formulas of Appendix E; the exhibited cycles — every elementary cube boundary together with three wrapping sheets — bound the kernel from below, and rank over $\mathbb{F}_p$ bounds it from above, since rank can only drop under reduction. The two bounds meet at $L = 1, \dots, 5$.

check: $\dim \mathbb{Z}_2 = L^3 + 2$ by rank, not by re-arranging the formula

check: cube boundaries and three wrapping sheets SPAN $\mathbb{Z}_2$

Remark 3. $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number; the Betti contribution is the final 3. The restriction $L \geq 3$ belongs to the twelve-neighbour Bloch adjacency, where $x+e$ and $x-e$ coincide at $L = 2$. The chain-level count has no such restriction and holds at $L = 1, 2$ as well.

check: the L^3+2 count is chain-level, not the Bloch convention

The following sum rule joins Theorems 1 and 2, which are otherwise two unrelated statements: a 3×3 spectrum and an $(L^3 + 2)$ -dimensional space.

Theorem 4 (Bloch-chain bridge). *On the periodic L^3 complex,*

$$\sum_k (3 - \text{rank } B(k)) = L^3 + 2,$$

the sum running over the L^3 allowed momenta, with nullity profile 3 at Γ exactly once and 1 at each of the other $L^3 - 1$ momenta.

Proof. $B(0) = 0$ contributes 3. For $k \neq 0$ some $d_j \neq 0$, and Theorem 1 gives $\text{rank } B(k) = 2$, contributing 1 each. Summing gives $3 + (L^3 - 1) = L^3 + 2$. $\square$

check: the Bloch and chain routes to the carrier agree

Remark 5. The two decompositions produce the same two numbers from different objects. The 3 is the triple degeneracy of $B(0) = 0$, not $b_2(T^3)$; the $L^3 - 1$ counts non- Γ momenta, not cubes. Their agreement is a consistency check on the whole construction, not a restatement.

3.1 The wrapping sheets are cycles, not harmonic

Theorem 2 exhibits its three H_2 generators explicitly as wrapping sheets $s_{(ij)} = \sum_{x: x_m=0} [x; i, j]$. These satisfy $\partial_2 s = 0$ exactly, so they are cycles and they span the quotient $\ker \partial_2 / \text{im } \partial_3$. They are *not* harmonic: $\partial_3^\dagger s \neq 0$, and

$$\frac{\langle s, \partial_3 \partial_3^\dagger s \rangle}{\|s\|^2} = 2 \quad \text{exactly, for every } L \geq 2. \quad (7)$$

check: FINDING: the wrapping sheets are cycles but NOT harmonic

The distinction matters because H_2 is used in two senses: the quotient in Theorem 2, which the sheets span, and the harmonic subspace $\ker \partial_2 \cap \ker \partial_3^\dagger$, which they do not lie in. Nothing downstream breaks —

Proposition 24 rests on $B^\dagger w = 0$, which covers all of $\ker \partial_2$ — but a real-space statement phrased for the harmonic subspace does not cover the generators this construction exhibits.

4 Exact all-rank dynamics at second order

Two plaquettes sharing a link leave six nonshared fundamental links after the external contractions. On the shared link the fusion channels depend on the relative induced orientation,

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}, \quad F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F, \quad (8)$$

with dimensions and quadratic Casimirs

$$\begin{aligned} d_{\mathbf{1}} &= 1, & C_{\mathbf{1}} &= 0, \\ d_{\text{Adj}} &= N^2 - 1, & C_{\text{Adj}} &= N, \\ d_{\Lambda^2 F} &= \frac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \frac{(N+1)(N-2)}{N}, \\ d_{\text{Sym}^2 F} &= \frac{N(N+1)}{2}, & C_{\text{Sym}^2 F} &= \frac{(N-1)(N+2)}{N}. \end{aligned} \quad (9)$$

The six nonshared half-links contribute $3C_F$ and the fused link $C_\rho/2$, against an external one-plaquette energy $2C_F$, so the resolvent denominator is $C_F + C_\rho/2$ exactly.

check: each channel gap is $C_F + C_\rho/2$, and the weights sum to one

4.1 The channel weights are a theorem

Earlier editions assigned the channel projector squared norm d_ρ/N^2 by an isotropy hypothesis, and every statement in this section rests on that assignment. It is not a hypothesis.

Theorem 6 (Shared-link channel weights). *In the normalised two-plaquette state, the weight of the shared-link channel ρ is exactly d_ρ/N^2 , for both orientation families and every $N \geq 3$.*

Proof. Two steps. First the six nonshared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar, so the amplitude is $\text{Tr}(AU) \text{Tr}(BU^{\pm 1})$ with A, B independent Haar and U the shared link. Integrating A and B contributes δ/N each and leaves a pure degree- $(2, 2)$ moment of U .

Second, two such moments settle both families. In the Weingarten calculus, whose asymptotic origin is [3] and whose finite- N closed forms are [9], the order-two values are $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2 - 1))$ — the inverse of the S_2 Gram matrix $\begin{pmatrix} N^2 & N \\ N & N^2 \end{pmatrix}$ —

the index sums give

$$\begin{aligned} M_{\text{direct}} &= \sum_{ijkl} \langle |U_{ij}|^2 |U_{kl}|^2 \rangle = N^2, \\ M_{\text{cross}} &= \sum_{ijkl} \langle U_{ij} U_{kl} \bar{U}_{il} \bar{U}_{kj} \rangle = N. \end{aligned} \quad (10)$$

For the like family, $\text{Tr}(AU)\text{Tr}(BU)$ decomposes into $\text{Sym}^2 \oplus \Lambda^2$ by symmetrising the two index pairs jointly, and the two squared norms are $(M_{\text{direct}} \pm M_{\text{cross}})/2$. Normalising,

$$n_{\text{Sym}^2} = \frac{N+1}{2N} = \frac{d_{\text{Sym}^2 F}}{N^2}, \quad n_{\Lambda^2} = \frac{N-1}{2N} = \frac{d_{\Lambda^2 F}}{N^2}.$$

For the mixed family the singlet component of $U_{ij}\bar{U}_{lk}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1 against $M_{\text{direct}} = N^2$, so $n_1 = 1/N^2 = d_1/N^2$ and $n_{\text{Adj}} = 1 - 1/N^2 = d_{\text{Adj}}/N^2$. $\square$

check: the shared-link weights are Weingarten, not an isotropy assumption

Remark 7. The derivation imports nothing: the Weingarten pair is the inverse of the S_2 Gram matrix and the index sums in (10) are explicit, so the reader can check it without leaving this page. That the pair is not two fitted constants is separately checked — the identity-permutation value times $N^2 - 1$ is 1 identically in N , so the $SU(3)$ numbers are a specialisation of a rank-generic formula. Together these are why the all-rank second-order result is self-contained rather than conditional. Appendix B records the index sums.

check: the Weingarten route is independent of the corpus

Remark 8 (External corroboration). The same numerator appears in the quantum-simulation literature from an unrelated direction. Ciavarella, Burbano and Bauer [11] publish a finite-rank plaquette matrix element $|M_\rho|^2 = \dim \rho / (\dim A \dim R)$; for two fundamental factors $\dim A = \dim R = N$ and this is d_ρ/N^2 identically in N . That is provenance for a chain already checked here, not a second derivation, and it does not check the premise that four channels exhaust the shared-link routes.

check: the published dimension-ratio matrix element is this registry's weight formula

With Theorem 6 the resolvent weight is

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}, \quad (11)$$

and summing within the two orientation families,

$$A_N = w_1 + w_{\text{Adj}} = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad (12)$$

$$B_N = w_{\Lambda^2} + w_{\text{Sym}^2} = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (13)$$

check: the four channel weights follow from dimension and Casimir

check: A.N and B.N are the channel sums, not transcriptions

4.2 The shared-link law

Theorem 9 (All-rank shared-link law). *For every integer $N \geq 3$,*

$$H_{\text{eff},-}^{(N)}(k, u) = E_{\text{flat},N}(u)I + t_N u^2 B(k)B(k)^\dagger + O(u^3),$$

with $E_{\text{flat},N}$ momentum independent and

$$t_N = B_N - A_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0.$$

The three bands through second order are $E_{\text{flat},N}(u)$ and $E_{\text{flat},N}(u) + t_N u^2 q(k)$ twice.

Proof. The diagonal and within-plaquette terms are scalars in face-orbital space. Every surviving second-order inter-plaquette process uses one shared link — see Proposition 10 for why the routes that do not are absent from this sector — and its relative orientation is the incidence product appearing in $S = BB^\dagger - 4I$. The contributions (12) and (13) therefore assemble as $d_N I + t_N S$; absorbing $-4t_N$ into the scalar and applying Theorem 1 gives the bands. Positivity: for $N \geq 3$ every denominator factor is positive and $N^2 - 4 > 0$. $\square$

check: t.N = B.N - A.N

check: t.N > 0 for N >= 3

Two consequences,

$$t_3 = \frac{5}{612}, \quad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} - \frac{1021}{256N^9} + \dots \quad (14)$$

check: t.2 = 0 and t.3 = 5/612

check: large-N expansion of t.N through 1/N^9

4.3 The vacuum-mediated route

The proof of Theorem 9 asserts that only shared-link routes survive. That is false in general and true here, for a reason worth stating: it is a charge-conjugation selection rule, and omitting it is a recorded error, not a hypothetical one.

Proposition 10 (Vacuum-mediated route). *The route $\langle p'|V|0\rangle\langle 0|V|p\rangle/E_0$ is available for any pair of plaquettes, sharing a link or not. It vanishes identically in the charge-odd sector, because V is charge-even and $|0\rangle$ is charge-even, so $\langle 0|V|p,-\rangle = 0$. Consequently the charge-odd hopping $t_N = B_N - A_N$ carries no such*

term, while the charge-even shared-link element acquires $1/C_F$:

$$\ell_N = A_N + B_N + \frac{1}{C_F} = -\frac{2N(3N^2 - 5)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}.$$

check: `ell_N = A_N + B_N + 1/C_F, the vacuum-mediated route at every rank`

Remark 11. At $N = 3$, $A_3 + B_3 = -481/612$ and $1/C_F = 3/4$, giving $\ell_3 = -11/306$. The first of these is exactly the value once published for the charge-even hopping and later superseded; the correction is precisely the omitted route. Stating the selection rule turns a one-rank erratum into an all-rank formula.

One thing this proposition does *not* do is dispose of the route between non-adjacent pairs. Taken over all pairs it would be a rank-one operator supported at $k = 0$, which is not the twelve-neighbour adjacency ℓ_N multiplies; which contributions survive in H_2 is settled by the linked, vacuum-subtracted prescription of (5), not by anything derived here. What the check below establishes is the all-rank value ℓ_N of the shared-link element, and the disposition of the non-adjacent contributions is prescription rather than result.

4.4 Finite-volume width

Let $q_{\max}(L)$ be the largest sampled q on the periodic L^3 grid. Then

$$q_{\max}(L) = \begin{cases} 12, & L \text{ even,} \\ 12 \cos^2(\pi/2L), & L \text{ odd,} \end{cases} \quad (15)$$

because only at even L does some k_j reach π exactly. The width of the full retained charge-odd manifold is $W_{N,L}^{(-)} = q_{\max}(L) t_N u^2 + O(u^3)$, so for even L and in the thermodynamic limit $[u^2]W^{(-)} = 12t_N \sim 3/N^3$. This concerns the stiffness of the dispersive doublet, not motion of the lowest branch, which is flat at this order.

check: the zone maximum of q is 12 only at even L

The smallest nonzero incidence eigenvalue is $q_{\min} = 4 \sin^2(\pi/L)$, attained by one unit of momentum in a single direction.

check: `q_min on the L-torus grid is 4 sin^2(pi/L)`

4.5 The orientation signs are essential

Removing the orientation signs — from the entries of (6) and from the phases, so each $e^{ik} - 1$ becomes $e^{ik} + 1$ — gives an unsigned incidence $\mathcal{N}(k)$ with $A + 4I = \mathcal{N}\mathcal{N}^\dagger$. Its adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, which share no eigenvalue, so no level can be momentum independent. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry.

This control is not hypothetical: the unsigned spectrum is the charge-even sector's, which Section 6 computes in closed form, and the two spans are the two bandwidths. The signed span is $8 - (-4) = 12$, giving the charge-odd width $12t_- = 5/51$ at $N = 3$; the unsigned span is $12 - (-4) = 16$, giving the charge-even width $16|\ell_3| = 88/153$.

check: the two band spans ARE the two incidence spectra

5 The second-order coefficient on a two-cube space

Everything above derives t_N from a resolvent ledger indexed by the fusion channel of one shared link. That ledger is representation theory carried out on a single plaquette pair; it has never been tested against an operator calculation on a space large enough for two cubes to interact. This section reports such a test and, more importantly, states what it settles.

The construction is the open face-sharing $(3, 2, 2)$ prism with its eleven oriented plaquettes, in the exact-Casimir $B = 6$ $SU(3)$ truncation, of dimension 1,590,462. Its complete charge-odd one-plaquette shell at $E_* = 8/3$ has dimension eleven. Writing K_{LR} for the raw second-order operator and K_L, K_R, K_F for the left-cube, right-cube and shared-face operators in their own coordinates, the connected operator is defined by the literal Möbius fold

$$K_{\text{conn}} = K_{LR} - J_L K_L J_L^\dagger - J_R K_R J_R^\dagger + J_F K_F J_F^\dagger, \quad (16)$$

and the same fold applied to the geometry gives G_{conn} . The result is

$$K_{\text{conn}}^{(2)} = \frac{5}{612} G_{\text{conn}} + D, \quad (17)$$

with D diagonal. Because G_{conn} turns out to be four disjoint 2×2 blocks plus three isolated directions (Appendix F), the spectrum is closed form, $\{0, (-\frac{15}{4})^2, (-\frac{34}{9})^4, (-\frac{129}{34})^4\}$ — exact arithmetic on entries that are themselves reconstructions, a boundary made precise below. The delivery reports the coefficient recovered target-blind — supplied to no builder, graph fit, branching rule or validator, with the payload hashes frozen before any comparison — and this repository re-thresholds those gates rather than re-running the build, including a deliberately wrong-sign control that the held-out scaling rejects. What that establishes is runtime nonuse of the comparator on the delivered dependency path, which is not a preregistration.

check: the connected two-cube geometry has exactly four cross-cell pairs, each -1

check: the $B=6$ connected kernel is $(5/612) G_{\text{conn}} + \text{diag}$, with the certified spectrum

check: the certificate's own gates pass, target-blind, with the wrong-sign control rejected

Resolving (17) by the irreducible representation the shared link actually carries in the ranked intermediate state gives six coefficients rather than four, because a link distinguishes ρ from $\bar{\rho}$:

$$\begin{array}{c|cccccc} \rho & \mathbf{1} & \mathbf{3} & \bar{\mathbf{3}} & \mathbf{6} & \bar{\mathbf{6}} & \mathbf{8} \\ \hline c_\rho & \frac{1}{12} & -\frac{1}{12} & -\frac{1}{12} & -\frac{1}{9} & -\frac{1}{9} & \frac{16}{51} \end{array} \quad (18)$$

check: the B=6 six-channel census sums to the registry's own t.3 = 5/612

That these six sum to 5/612 is weak evidence: six rationals reach one target in many ways, and a fitted set would pass it. The statement worth making is stronger, and it is what makes this more than a coincidence of one sum. First, though, the boundary.

The six coefficients of (18) are not symbolic: the delivery stores them as rational reconstructions of pinned finite-precision Clebsch–Gordan contractions, with a maximum reconstruction residual of 2.1×10^{-14} and its own certificate declaring `symbolic_exact_local_amplitudes: false`. What follows is therefore reported, conditional on those six numbers, and not a symbolic identity in the sense of Theorem 6.

Reported computational result 12 (The census is the Weingarten ledger). At $N = 3$ the six reconstructed link-irrep coefficients of (18) are the four fusion-channel weights (11) individually:

$$\begin{aligned} c_1 &= -w_1, & c_8 &= -w_{\text{Adj}}, \\ c_3 &= c_{\bar{3}} = \frac{1}{2}w_{\Lambda^2 F}, & c_6 &= c_{\bar{6}} = \frac{1}{2}w_{\text{Sym}^2 F}. \end{aligned}$$

Substituting (9) into (11) at $N = 3$ gives $w_1 = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$ and $w_{\text{Sym}^2 F} = -2/9$; comparison with (18) is then arithmetic. Six reconstructed rationals landing on four predicted ones, with the two coincidences the correspondence requires, is a strong consistency statement and not a proof that the underlying contraction is exact.

check: the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel

The correspondence is scoped to what the Casimir closure exhausts: adjacent, one-action, shared-link intermediates at order u^2 . Same-face diagonal routes, on-site terms and higher-action intermediates lie outside it — the $B = 6$ scalar below is exactly a place where one of them is missing — and so does any untruncated-Hilbert-space statement.

Two features of the correspondence carry the content. The mixed family enters negated because $t_N = B_N - A_N$, which is the same relative sign the incidence orientation carries. And each like-family *fusion* weight is split evenly between the irrep and its conjugate —

the two orientations of the shared link weighted equally — which is measured here on the 1,590,462-state space rather than assumed.

Conjecture 13 (All-rank even split). For every $N \geq 3$ each like-family fusion weight splits evenly between an irrep and its conjugate in the link-resolved census.

This is a conjecture and nothing here promotes it. Falsifying it means a link-resolved census at some $N \geq 4$ in a truncation that still retains every adjacent shared-link channel; at $N = 4$ the endpoint budgets make that $B \geq 33/4$, and no such construction has been run.

5.1 One truncation, two constructions

The two-cube space also isolates why a cutoff can reverse the sign of t_3 . An adjacent shared-link channel ρ is reachable only if the endpoint Casimir budget $C_2(\rho) + 2C_2(\mathbf{3})$ fits inside the truncation: the sextet needs exactly 6 and the adjoint 17/3, so both survive $B = 6$ and neither survives $B = 4$.

check: B=6 retains every adjacent shared-link channel; B=4 provably cannot

Corollary 14 (What a link cutoff drops). *Retaining only the singlet and $\Lambda^2 F$ routes projects the second-order hopping to*

$$w_{\Lambda^2 F} - w_1 = -\frac{1}{12},$$

opposite in sign to t_3 and 10.2 times larger in magnitude. What is dropped is

$$w_{\text{Sym}^2 F} - w_{\text{Adj}} = \frac{14}{153},$$

and $-\frac{1}{12} + \frac{14}{153} = \frac{5}{612} = t_3$. This is arithmetic on (11) alone.

The published $p + q \leq 1$ link cutoff retains exactly those two routes, so $-1/12$ is its projected hopping and $14/153$ is the counterterm that completes it — not 5/612, which would double-count what it already carries. Conditional on (18), the $B = 4$ truncation of the two-cube space retains the same two fusion channels, split by link orientation, and therefore lands on the same two numbers: $c_1 + c_3 + c_{\bar{3}} = -1/12$ and $c_6 + c_{\bar{6}} + c_8 = 14/153$.

check: the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel

check: FINDING: the T1 link cutoff reverses the sign of t.3, and 14/153 is what it omits

Remark 15. Two single-cube reconstructions bracket the same statement. A full $B = 4$ open cube assembled from the published `ymcirc` $d = 3$, $B = 4$ plaquette table [12] reproduces $-1/12$ and the truncated shell ordering to 4.3×10^{-11} ; a $B = 6$ cube reproduces $+5/612$

and the inverse ordering to 8.4×10^{-15} . Both locate the missing weight at the same 14/153.

Neither is an independent replication. Both read tabulated plaquette matrix elements of the same lineage, and the published closed form [11] is d_ρ/N^2 — the very numerator of (11). This is a cross-check of assembly, not a third confirmation of t_3 ; the exact rational identity is Corollary 14, which needs no tolerance. The $p + q \leq 1$ nomenclature and the finite-rank matrix element are both from [11], and single-cube $SU(N)$ diagonalisation in this style goes back to [8].

A counterterm added to a published T_1 Hamiltonian is therefore 14/153 and not 5/612, which would double-count the two routes T_1 already carries. One coincidence of value is worth naming: $-1/12$ is also the singlet weight w_1 , and the cutoff value is a *difference* of two weights, not a weight.

check: FINDING: a full T1 = B = 4 cube Hamiltonian reproduces -1/12 and the reversed shell

check: FINDING: the B = 6 cube flips the sign back to +5/612, closing the decisive test

The $B = 6$ cube is channel-complete for the hopping and *not* for the scalar: its scalar is 39/68 against the bridge's 11/34, differing by exactly 1/4, the local same-face sextet route, which first enters at cutoff 7.

check: the B = 6 scalar misses the bridge's by exactly the same-face sextet route

The finite-volume fingerprint of the same certificate is worth recording because its most discriminating multiplicity is not a feature of the truncation: on the periodic 3^3 torus the incidence spectrum is $\{0^{29}, 3^{12}, 6^{24}, 9^{16}\}$, and $29 = L^3 + 2$ is Theorem 2.

check: the certificate's finite-volume fingerprints, and $29 = L^3 + 2$ is the Lean cycle count

Remark 16 (Scope). Everything in this section is second order in u , finite volume, strong coupling, one sector. It bears on t_N and decides nothing about the fourth-order coefficient of Section 9. The 1,590,462-state construction was not re-run here: its arithmetic is audited against geometry rebuilt independently, and its diagonal D is B -truncated and not proved cutoff-stable. The held-out finite- u validation that rejects the wrong-sign control is a 66-dimensional $P + Q_1$ word space, not the full Hamiltonian, and no radius-three stability claim was executed. No external group has reproduced the release.

6 The charge-even sector in closed form

The unsigned incidence $\mathcal{N}(k)$ of Section 4 is not a hypothetical control: it is the charge-even sector's Bloch

symbol. Write

$$\hat{a}_m = 2 + 2 \cos k_m = |1 + e^{ik_m}|^2, \quad p = \sum_m \hat{a}_m.$$

The hat is not decoration: $\hat{a}_m$ is the *complement* of the $a_m = 4 \sin^2(k_m/2)$ used in the charge-odd shape basis (20) below, $\hat{a}_m + a_m = 4$, and that complementarity is the sum rule below.

Theorem 17 (Charge-even Bloch cubic). $p + q = 12$ pointwise, and the characteristic polynomial of $\mathcal{NN}^\dagger$ is

$$\mu^3 - 2p\mu^2 + p^2\mu - 4\hat{a}_1\hat{a}_2\hat{a}_3 = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3,$$

with the charge-even adjacency eigenvalues $\lambda = \mu - 4$. The charge-odd counterpart of the same computation is $\mu(\mu - q)^2$.

check: the C-even characteristic polynomial is $\mu(\mu - p)^2 - 4 \hat{a}_1 \hat{a}_2 \hat{a}_3$

check: $p + q = 12$: one zone function runs both sectors

The closed form is complete rather than merely consistent at sampled momenta: the characteristic polynomial of the 81×81 and 192×192 finite plaquette adjacencies factorises coefficient for coefficient over $\mathbb{Z}$ into this cubic over the momentum grid.

check: the Bloch cubic IS the finite $L = 3$ and $L = 4$ plaquette spectrum, exactly

Proposition 18 (Range and attainment). $\lambda \in [-4, 12]$ exactly. The top $\lambda = 12$ is attained only at Γ ($\hat{a}_1 = \hat{a}_2 = \hat{a}_3 = 4$). The bottom $\lambda = -4$ is attained on the whole of the three planes $k_j = \pi$, where the constant term vanishes, and only at R is it a triple root.

Proof. Write $f(\mu) = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3$. Then $f(0) = -4\hat{a}_1\hat{a}_2\hat{a}_3 \leq 0$ and $\mu(\mu - p)^2 < 0$ for $\mu < 0$, so no root lies below 0; and $f'(\mu) = (3\mu - p)(\mu - p) > 0$ for $\mu > p$, with $f(16) = 16(16 - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3 \geq 0$ because $p \leq 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 \leq 64$. Hence $\mu \in [0, 16]$ and $\lambda = \mu - 4 \in [-4, 12]$. Equality at the top forces $p = 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 = 64$, which is Γ . Equality at the bottom is $f(0) = 0$, i.e. $\hat{a}_1\hat{a}_2\hat{a}_3 = 0$, i.e. some $k_j = \pi$; there $f(\mu) = \mu(\mu - p)^2$ and the root $\mu = 0$ is simple unless $p = 0$, which is R alone. $\square$

check: the C-even range is exact; the top is attained only at Gamma, the floor on three planes

check: the C-even band touches its floor exactly on the three planes $k_j = \pi$

Remark 19. A flat charge-even branch would need $f(\mu_0) = 0$ identically in k , which the constant term already forbids. The charge-even kernel is three measure-zero planes against the charge-odd kernel's whole band:

the two sectors are two readings of one geometric object, and only the signed one is flat.

check: the C-even band touches its floor exactly on the three planes $k_j = \pi$

check: the C-even spectra at the four high-symmetry momenta

6.1 One assembly, both sectors, both orders

The $r = 2$ content below is all-rank and checked. The $r = 3$ values — $t_{3,+} = -6335/249696$ and everything computed from it — come from the same supplied domino ledger as Section 7, and are conditional on it in the same way.

The scalar ledger is not a list of separate facts. Writing λ for the adjacency eigenvalue and $\text{tower}_{r,s}$ for the within-plaquette term in canonical u ,

$$E_s(\lambda, r) = \text{tower}_{r,s} + 12 \text{leak}_{r,s} + \lambda t_{r,s}, \quad (19)$$

with $\lambda \in \{-4, 8\}$ in the charge-odd sector (carrier and dispersive top), $\lambda \in \{12, -4\}$ in the charge-even sector, and $\lambda = 0$ returning the on-site diagonals. This reproduces every registered diagonal and band value — nine of them — across both sectors and both orders. The tower term is the certified coupling conversion $4\Delta(3u/2)$, so (19) takes no unregistered input. The twelve is a count on this lattice, not a citation: the plaquette graph is 12-regular and two distinct faces share at most one link.

check: one assembly formula gives every registered band value

check: one assembly formula gives every C-even value at both orders

check: the plaquette graph is 12-regular and two faces share at most one link

Because $\text{leak}_{r,+} = t_{r,+}$ at both computed orders, the whole charge-even sector collapses to $\text{tower}_r + (12 + \lambda)t_{r,+}$, a one-parameter family. That equality is verified and unexplained: at third order the charge-odd leakage separates from the charge-odd hopping while the charge-even identity survives. It is recorded as a unifying candidate with a falsifier — an order at which the two are computed independently and differ — not as a mechanism.

check: both declared coincidences, checked: one is `ell_N` at all ranks, the other is bare

6.2 Where the rest frame is not blind

Proposition 20 (Charge-even Γ splitting). *At Γ the charge-even spectrum is $\{12, 0, 0\}$, so the A_1^{++} level and*

the E^{++} doublet are distinct, and

$$E(A_1^{++}) - E(E^{++}) = 12 t_{r,+}$$

exactly: $-22/51$ at $r = 2$ and $-6335/20808$ at $r = 3$.

check: the C-even Gamma point pins t_+ , exactly where no C-odd Gamma datum can

This is the structural counterpart of the charge-odd blindness recorded in Section 7. The signed spectrum at Γ is $\{-4, -4, -4\}$, one level, so no charge-odd rest-frame series can see the hopping; the unsigned spectrum has two levels, so the charge-even rest frame can. The A_1^{++} branch is isotropic at leading order without assuming it, with curvature $\frac{4}{3}|t_{r,+}|$ in all three directions — $22/459$ at second order and $6335/187272$ at third.

check: the C-even curvature is $(4/3)|t_+|$ at both orders, and isotropic

At both computed orders the charge-even bandwidth is $16|t_{r,+}|$ and the charge-odd manifold width $12|t_{r,-}|$. At any order at which the correction stays in the incidence algebra — which Proposition 24 warns is not automatic — the second-order ratio is $4(3N^2 - 5)/[3(N^2 - 4)]$, finite and greater than one at every rank $N \geq 3$, decreasing from $88/15$ at $N = 3$ towards 4. So at second order the charge-even band is the wider one at every rank, and that asymmetry is a channel statement, not a geometric one.

check: the C-even bandwidth is $16|t_+|$ at every order; the C-odd manifold width is $12|t_-|$

7 $SU(3)$ through third order

The *third-order* content of this section is conditional on supplied exact-arithmetic ledgers. Its second-order content is not: leak_2 is the all-rank ℓ_N of Proposition 10 at $N = 3$, and the tower is the certified coupling conversion. The within-plaquette charge-odd tower contributes $8/3 + u + u^2/2 + 7u^3/32$.

Reported computational result 21 (Third-order ledger). The supplied ledgers report, in exact rationals,

$$b_3 = \frac{1975}{124848}, \quad \text{leak}_3 = -\frac{12331}{249696},$$

and the assembled flat scalar $d_3 = \frac{7}{32} + 12 \text{leak}_3 - 4b_3 = -\frac{109151}{249696}$.

check: `d_3 = 7/32 + 12*leak_3 - 4*b_3`

check: `leak_3` is assembled from the domino diagonal and the vacuum piece

Conditional corollary 22 (Third-order factorisation). If the lifter census is exhaustive and the linked ledger complete, then

$$H_{\text{eff},-}(k, u) = E_{\text{flat}}(u)I + \tau(u)B(k)B(k)^\dagger + O(u^4),$$

$$E_{\text{flat}}(u) = \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3, \\ \tau(u) = \frac{5}{612}u^2 + \frac{1975}{124848}u^3.$$

check: `E.flat` and `t(u)` carry the ledger coefficients

The second-order assembly is $\frac{1}{2} + 12\text{leak}_2 - 4t_-$ with $\text{leak}_2 = -11/306$, giving the charge-odd diagonal $7/102$ and then the flat scalar $11/306$.

check: `d.- = 1/2 + 12*leak.2, and leak.2 = -11/306`

Remark 23 (What the external comparison can and cannot do). Hamer’s 1989 rest-frame series [7] agrees with E_{flat} at every shared order. It cannot test τ . The hopping enters the spectrum only through $\tau(u)q(k)$ and $q(0) = 0$, so every *charge-odd* Γ -point datum is blind to it — the charge-even rest frame is not, by Proposition 20; the one-parameter family $(b_3 + \varepsilon, \text{leak}_3 + \varepsilon/3)$ leaves d_3 and therefore the entire comparison unchanged. What the agreement pins is the combination $12\text{leak}_3 - 4b_3$. The $\lambda = 8$ band top separates them, via (19). Proposition 20 is the charge-*even* analogue and separates leak_+ from t_+ ; it says nothing about b_3 or leak_3 .

check: FINDING: no Gamma-point datum can constrain the hopping

8 What the homology protects

Proposition 24 (Boundary-factorised rigidity). *If an order- r Bloch correction has the form $H_r(k) = a_r I + b_r S(k) + B(k)M_r(k)B(k)^\dagger$, then for every $k \neq 0$, $H_r(k)w(k) = (a_r - 4b_r)w(k)$: the correction shifts the carrier without dispersing it.*

Proof. $B^\dagger w = 0$ and $Sw = -4w$, both from Theorem 1. $\square$

check: `B B~dagger = q I - d conj(d)~T` for the curl incidence

In real space this is the orthogonality of the two cellular Laplacians, $L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0$, so any polynomial in them acts on $\ker L_2^\downarrow \cap \ker L_2^\uparrow$ by its constant term. The proposition is not an unconditional all-orders statement: $\{BMB^\dagger\}$ is not a two-sided ideal in $\text{End}(C_2)$, and a general higher-order correction need not factor through links. Cubic symmetry alone permits carrier-projected fourth-order shapes built from

$$1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \quad (20)$$

with $e_2 = \sum_{i < j} a_i a_j$, $e_3 = a_1 a_2 a_3$ and $a_i = 4 \sin^2(k_i/2)$. The $1/q$ is the carrier projection: $\|w\|^2 = q$ by Theorem 1, so the carrier-projected correction is $\varepsilon_4 = (w^\dagger H_4 w)/q$ and the numerator, not ε_4 , is the polynomial.

check: the carrier projection is where the $1/q$ comes from

check: clearing the denominator reproduces the five-element numerator basis

9 The fourth-order boundary

Two supplied fourth-order records agree on the axial datum and on the appearance of mobility, and disagree on one off-axis mixed-gradient coefficient C_{shp} . This edition does not adjudicate them, and no fourth-order band or bandwidth is used above. What it does is state precisely where the disagreement lives, because that geography is now checked and it is what a decisive calculation must target.

The two records do agree on the axial datum, but the corpus’s own quoted tolerance does not cover that agreement: the α gap is 2.43×10^{-13} against a stated bound of 2.3×10^{-13} . The sealed core holds internally — α_{new} and $4A_{\text{new}}$ agree to 2×10^{-15} — which is consistent with the quoted bound being too tight, but two float comparisons do not discriminate that from a shared upstream slip. Recorded as a discrepancy rather than resolved, and the bound is not widened.

check: `v10a.26 A, B, D` match the sealed rationals within $2.3\text{e-}13$

check: FINDING: `alpha_new` falls outside the corpus’s own $2.3\text{e-}13$ bound

9.1 The obstruction is polynomial

Theorem 25 (Obstruction certificate). *The two recorded off-axis coefficients enter (20) only through the shape $4e_2/q$, so the two records’ carrier numerators differ by $4\Delta_C e_2$ and their bands by $\Delta_C (4e_2/q)$. On any one-dimensional axial cut two of the three a_i vanish, so e_2 is the zero polynomial there and both differences vanish identically. Consequently no Γ -point or axial datum distinguishes the two records, at any precision. Off axis they do differ: $e_2(M) = 16$ and $e_2(R) = 48$, so the numerators differ by $64\Delta_C$ and $192\Delta_C$ and the bands by $8\Delta_C$ and $16\Delta_C$.*

check: FINDING: the retained Gamma/axis data cannot identify C_{shp}

check: `Phi_C` vanishes at Gamma along every direction

Remark 26. The vanishing is polynomial, not numerical: this is non-identifiability, not a limitation of the

data in hand. Re-anchoring cannot substitute, and neither record is preferred here. An off-axis contraction is the only decider.

check: the crosswalk is exactly scalar on the momentum axes

On the axial cut itself the invariants collapse to $q = L(k) = 4\sin^2(k/2)$ with $e_2 = e_3 = 0$, and dividing the raw cube-boundary numerator $(\alpha_N/4)L^2$ by $\|w\|^2 = q = L$ leaves a single power of L with coefficient $\alpha_3/4 = 5/48$.

check: on an axial cut the mixed invariants vanish and the norm divides

9.2 A second witness

Non-identifiability is better exhibited than argued from a difference between two records, since a difference could always be a computational error in one of them. The witness below is not itself a candidate for the physical coefficient — it is the balanced ϵ -free continuation of the all-rank shape family, and ϵ -free is precisely what the physical value is not. Its job is to make the $4\Delta_{C^2}$ family concrete with an exactly known shift.

Proposition 27 (Explicit second witness). *There is a value C_{alt} with $C_{\text{alt}} - C_{\text{old}} = 25/1024$ exactly, appearing verbatim in the pinned source edition, whose induced band separations are 0 at X , $25/128$ at M and $25/64$ at R . It is identical to C_{old} on every retained datum and distinct from it off-axis.*

check: FINDING: an explicit second witness C_{alt} exhibits the C^2 non-identifiability

check: the C_{alt} witness IS the balanced ϵ s-free continuation, and $\Delta_{C/(A/2)} = 15/32$

9.3 Where the coefficient lives

The four checkpoint deltas in the shape basis are

$$\begin{aligned}\Delta_X &= 4A, \\ \Delta_M &= 8A + 16B + 8C, \\ \Delta_P &= 6A + 8B + \frac{16}{3}C, \\ \Delta_R &= 12A + 48B + 16C + \frac{16}{3}D,\end{aligned}\tag{21}$$

and they invert exactly, so the four shape coefficients are recoverable from four checkpoints. Δ_X contains A alone, which is why the axial cuts agree whatever the dispute does.

check: checkpoint values at X, M, P, R

check: the four extraction formulas invert the ansatz

check: X is blind to B, C, D | it fixes A alone

Four further measurements locate the coefficient and bound where a disagreement could live. The distinction matters and is easy to lose: showing where a coefficient's *value* is concentrated is not by itself showing where a *discrepancy* can sit. Only the third item below does the second thing.

1. **The shape fit's C row sums to zero**, so no error in the Γ anchoring can move C_{shp} at all; the recorded run's own re-anchoring moved it by 4.6×10^{-15} , as the zero row sum requires. A scalar shift is translation-local and changes nothing observable, which is why the two anchored scalars $q_{\text{band}}^{(4)}$ and $m_{\Gamma}^{(4)}$ are differently anchored coordinates and not rival estimates.
2. **The coefficient's value is concentrated, not spread.** Six of the 189 kernel records — the normal $(0, 0, 1)$ shell — carry 81% of the total weight, while the 120 rotation records carry 6% and are 255 times less sensitive per record. This says where the coefficient lives, not yet where the two records can differ.
3. **The agreed axial coefficient pins the normal sector.** All six normal records share one amplitude h , and it sets both shapes rigidly: $A_{\text{normal}} = -h$ while the raw normal amplitude is $2h$, which in the shape normalisation of (20) is $C_{\text{normal}} = h/2$, so

$$C_{\text{normal}} = -\frac{1}{2}A_{\text{normal}} \text{ exactly.}$$

(The factor of four between $2h$ and $h/2$ is the raw-to-shape conversion, not a discrepancy.) Granting $A = 5/48$ — which both records share — together with the block decomposition, which puts $5/48 + 4x$ on the normal block and $-4x$ on the mixed one, therefore fixes h and with it 81% of the coefficient; and everything free to move without disturbing A totals less than half the disputed gap. A rival kernel must differ *structurally* in the A -carrying sector; no redistribution reaches the gap.

4. **The two kernels agree almost everywhere.** Their supports are identical and 144 of 189 records ride one common scale; the divergent cross sector is a single uniform ratio. The dispute is three amplitudes.

check: the shape fit's C row sums to zero, so no Gamma-anchor error can move C_{shp} at all

check: a translation-local scalar shift changes nothing observable

check: FINDING: C_{shp} is carried by 6 of 189 records, not spread across the kernel

check: FINDING: $A = 5/48$ pins the normal sector's whole C^4 contribution

check: C.normal = -A.normal/2: the agreed axial coefficient pins the normal channel

check: the two 189-record kernels agree everywhere except three amplitudes, and the on-site anchor swap moves C by exactly zero

Read together these four sharpen the target and prefer no side of it. They say that a rival kernel must differ structurally in the A -carrying sector rather than by redistribution, which is a constraint on *any* candidate, including both recorded ones. Neither record is promoted here and neither is withdrawn: the interval of *values* between them is not narrowed. What is narrowed is the class of *kernels* that can reach either end, and that constraint bears on both records alike.

9.4 What cannot decide it

A large sealed sweep over 609 marked clusters has been the standing candidate for settling this. It cannot: its engine assembles coefficients through a Γ -point scalar and emits no kernel block at all, and by Theorem 25 Γ -point data are structurally incapable of constraining C_{shp} . This is a statement about the instrument, made by reading it, not a report of a failed run.

check: FINDING: the marked-cluster engine emits the Gamma scalar only | a completed 609-sweep cannot decide C_{shp}

9.5 The tier collapse, as two integers

The two shape coefficients B and D vanish in the recorded kernel even though the two-hop enumeration generates both, and the degree count does *not* forbid them. The carrier numerator reaches a -degree three, because the projection contributes one further power of a : for any vector whose components have squared modulus a_i — the carrier is one, in whichever of the two equivalent labellings — the quadratic form $\text{diag}(a_i^2)$ evaluates to $\sum_i a_i^3 = q^3 - 3qe_2 + 3e_3$, an explicit witness carrying a nonzero e_3 . An earlier degree-bound mechanism proposed for the collapse was therefore retracted. What is checked is finer: every degree-three block amplitude is an integer multiple of one raw record weight x , and

$$B = 0 : (+1, +1, -2) \cdot x, \quad D = 0 : (-3, +6, -3) \cdot x$$

over the displacement shells. The kernel does populate the forbidden tier; the dynamics cancels it with small integers. Any mechanism must reproduce those two vectors.

check: RETRACTED: the vertex count does NOT forbid B_{shp} and D_{shp}

check: FINDING: the tier collapse is two integer cancellations at record level

check: the vanishing coefficients are exactly the degree-3 ones

9.6 A near- Γ statement that survives the dispute

One quantitative fourth-order statement can be made without adjudicating anything, because it depends on the kernel only through $\sqrt{W_4}$. With Jordan's inequality $q(k) \geq (4/\pi^2)|k|^2$ on the whole zone — tight at $k = 0$ and at the zone corner — and $\tau(u) \geq t_3 u^2$ for $u > 0$, the fourth-order shape terms are dominated by the second-order dispersion outside a ball of radius Ku with $K = (\pi/2)\sqrt{W_4/(\theta t_3)}$. The two records give $K = 17.04$ and $K = 23.66$ — floating point, since one of the two bandwidths exists only as a float — and the larger bounds both, so the exclusion radius can be stated now. The statement is non-vacuous only for $u < 0.133$ at $\theta = 1/2$, and the excluded zone fraction grows as u^3 .

check: Jordan bound $q(k) \geq (4/\pi^2)|k|^2$ holds on the whole zone

check: the criterion survives C2: K depends on the kernel only through $\sqrt{W_4}$

check: the statement is non-vacuous only below an explicit coupling

10 External comparisons

External results count here because they are *independent* — produced without knowledge of this program — not because they are published. Each is an assertion until something checks it.

Hamer 1989, both sectors. The 1^{+-} rest-frame series agrees with E_{flat} at every shared order: $16/3 \div 2 = 8/3$ exactly, $a_1 = +1$, and $2a_2, 4a_3$ matching $11/306$ and d_3 to 9.9×10^{-14} and 1.1×10^{-12} . The 0^{++} series matches the charge-even A_1^{++} Γ -point coefficients at x^2 and x^3 , with the sign structure ($a_1 = -1$ in the scalar channel where the carrier's is $+1$) reproduced; the E^{++} doublet is not constrained by it, which is why Proposition 20 needs the splitting and not the level. The conversion $m_n = 2^{n-1}a_n$ is $x = 2u$ together with Hamer's normalisation $ma = (g^2/2)M$ — the 2^n from the substitution, the half from the normalisation — exactly, with no 4^r ambiguity anywhere. Hamer states in print that his x^3 and x^4 terms disagree with the earlier series of [2]; that paper has not been read here, so the disagreement is recorded and not adjudicated.

check: Hamer's 1^{+-} series matches the C-odd Gamma-point coefficients through x^3

check: Hamer's 0^{++} series matches the C-even Gamma-point coefficients through x^3

check: the $m.n = 2^{n-1}a_n$ bridge is the $x = 2u$ conversion

check: the Hamer table is pinned, and the a.4 agreement is primary-source

The string tension. The Kogut–Pearson–Shigemitsu $SU(3)$ table [4] — read here from the 1980 preprint, the journal pages themselves being unread — equals this program’s certified σ series exactly, $2^{n-1}t_n = \sigma_n$ for $n = 2, \dots, 5$, in the physical sign convention $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$. The same series reproduces E_{flat} term by term through u^3 , including the discriminating u^3 coefficient.

check: the KPS 1980 string-tension table equals the certified sigma series EXACTLY

check: $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$, and C5 is the $n = 3$ case

check: the ratio and sigma series reproduce E_{flat} exactly

Euclidean series. Two printings of the same Münster/Seo series — the NPB 205 (1982) erratum row and Smit’s Table 1 scalar column, which credits them — agree rational for rational through u^8 . That is agreement transcription for transcription, not two independent computations, and it is recorded as such. One shift is recorded and not adjudicated: Münster’s 1985 table [5] differs from the 1982 erratum row at eighth order by exactly -96 , a fourth correction already in print by 1983 that no erratum records. This is a Euclidean record; no Hamiltonian comparison is drawn from it, and none is permitted.

check: the errata-resolved Euclidean series is doubly sourced, transcription for transcription

check: FINDING: Munster’s 1985 table shifts his 1982 erratum at eighth order

The overlap obstruction. Schierholz [6] measured in 1988 that at $\beta = 5.9$ a bare local operator carries only a couple of percent of the physical glueball state, and gave the scaling a^5 for a local dimension-four operator — which is why that percentage needs its coupling quoted alongside it. That makes the smeared basis structural rather than a convenience: better statistics cannot recover a power law. It is the published, independent form of the obstruction this program records as its own largest debt.

check: the overlap obstruction was published in 1988, and it scales

Cross-regime discipline. Two indexed papers declare an explicit scope firewall and neither supplies a value here: Davoudi–Raychowdhury–Shaw [13], whose

Hilbert-space cost comparison is locked to 1+1 dimensions with matter, and Kadam–Raychowdhury–Stryker [10]. The rule is enforced by the literature validator rather than recorded in prose.

check: a cross-regime paper never supplies a value

11 Scope

This work does not establish a four-dimensional infinite-volume Yang–Mills mass gap, a continuum glueball mass, convergence of the strong-coupling series to the continuum, regulator independence of the first dispersive order, the full $SU(3)$ fourth-order rest coefficient or off-axis shape, or an identification of any level here with a physical glueball. At Γ the three face orbitals transform as an axial vector of the cubic group, which with parity even and charge conjugation odd gives the retained-space label T_1^{+-} ; this is an operator-space assignment, not a spectral one.

The carrier is macroscopically degenerate at finite L , but its nearest incidence separation is $\Delta_L(u) = 4\tau(u) \sin^2(\pi/L) + O(u^4)$, which falls as L^{-2} and so provides no volume-uniform spectral isolation.

check: $\Delta_L = 4 \tau(u) \sin^2(\pi/L)$ is positive and falls as L^{-2}

Three crossings are named explicitly, and none of them is made here.

- **Finite lattice to infinite volume** requires an inhomogeneous Wilson free-energy bound with a useful $\log K_\alpha$ and a source-radius reduction. Both are open, and are carried in the accompanying gap register as G17.
- **Protected level to particle** requires a volume-uniform overlap of a smeared, dressed T_1^{+-} operator built on the carrier with the transfer-matrix spectrum, and multi-plaquette survival of the protected level (G18). The published overlap measurement of Section 10 is why this theorem must live in the smeared basis.
- **Lattice to continuum** (G19) is untouched. Small u is strong coupling; the continuum is the opposite end of the axis.

The second and third are the only route from the statements above to any sentence containing the words “mass gap”, and nothing in this edition shortens that route.

Two premises earlier in the paper are arguments in prose rather than results, and both are load-bearing. That the range of P_- is exactly the charge-odd one-plaquette span for $L \geq 3$ is argued in Section 3 from the link count of a reduced fundamental loop. That the four fusion channels exhaust the second-order shared-link routes is assumed in Theorem 9; Theorem 6 fixes

the weight of each channel and does not establish that the list is complete. Neither has a check.

12 Reproducibility

A portable standard-library Python program, `paper/verify_core.py` in the accompanying repository (the root-level file of the same name belongs to the earlier flat-band manuscript and is a different, smaller program), checks nine claim groups: the rational coefficient ledger; the Weingarten derivation of Theorem 6 by explicit index summation; the incidence identity, *decided* in exact Gaussian-rational arithmetic at rational torus points rather than checked to a residual; the chain condition $\partial_2\partial_3 = 0$ over $\mathbb{Z}$ and the finite-torus ranks with an explicit kernel basis; the Rayleigh quotient of the wrapping sheets; the two-cube connected geometry, its exact $B = 6$ and $B = 4$ spectra, and Theorem 12 channel by channel; the charge-even cubic of Theorem 17 with $p + q = 12$ and the four high-symmetry spectra; the checkpoint algebra of (21) together with the vanishing of e_2 on every axial cut; and the coupling convention. Every value in it is a **Fraction**: no float is constructed anywhere, so “agreement” there always means equality. It has no dependencies and runs in about half a second.

Nothing in `verify_core.py` assumes or decides a value of C_{shp} . Its fourth-order group is the obstruction as arithmetic: the extraction formulas, the blindness of X , and the identical vanishing of e_2 on axial cuts.

Every theorem, proposition, corollary and reported result above names the machine check that establishes it, and the appendix tables name checks too. Two displayed objects deliberately carry none, because neither is a result of this program: the dimension and Casimir table (9), which is standard $SU(N)$ representation theory, and the boundary formulas of Appendix E, which are definitions — what is checked there is that they compose to zero. The checks live in the accompanying repository, where `workhouse verify --only '<name>'` reruns any one of them alone, with its numbers, in about a second. That device is itself checked: a registered invariant resolves every printed label against the live registry and fails if one is renamed or deleted, so the paper cannot go on printing a command that no longer resolves.

This edition pins commit `f25328f8d6658af63588fb3d30dbd3b7f6ede9c0` of the accompanying repository, at which it carries the counters in Table 1. They describe *that* commit, and each row has its own command: after `git checkout f25328f`, `make verify` gives the T_1 and T_2 rows, `make lean` the T_0 row, and `make check` the test count. They are provenance rather than independent evidence; the argument is above, and the checks are the checks.

Layer	Count
T_0 — Lean 4 theorems, no sorry	40
T_1 — exact re-derivations	179
T_2 — numerical, within a stated tolerance	46
Repository tests	472

Table 1: Counters at the pinned commit. Every T_0 theorem’s axiom footprint lies in `{propext, Classical.choice, Quot.sound}`: thirty-nine use all three and `dim_Z2` uses `propext` alone.

A smaller layer is proof-checked rather than merely re-derived. The Lean development compiles the rank law with denominators cleared, $t_3 = 5/612$ and $t_2 = 0$, the third-order ledger identity, the axial law at every exceptional rank, the cycle count $(L^3 - 1) + 3 = L^3 + 2$, the four checkpoint-extraction formulas and the deltas they invert, and four statements about the charge-even cubic including the identity that its middle coefficient is p^2 . It is exact-rational and polynomial algebra only: no perturbative derivation, no operator theory, and nothing that bears on C_{shp} appears there, because none of that reduces to rational arithmetic.

Machine checks certify algebra relative to stated definitions and inputs; they do not replace a derivation, and they do not certify an uncompleted off-axis contraction — which is why Section 9 stops where it does.

A Channel weights in closed form

Substituting (9) into (11),

$$w_1 = -\frac{2}{N(N^2 - 1)}, \quad w_{\Lambda^2 F} = -\frac{N - 1}{(N + 1)(2N - 3)},$$

$$w_{\text{Adj}} = -\frac{2(N^2 - 1)}{N(2N^2 - 1)}, \quad w_{\text{Sym}^2 F} = -\frac{N + 1}{(N - 1)(2N + 3)}.$$

Their family sums are (12) and (13), and the difference has numerator $2N(N^2 - 4)$, which is the $SU(2)$ zero and the positivity for $N \geq 3$ at once.

check: the four channel weights follow from dimension and Casimir

At $N = 3$: $w_1 = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$, $w_{\text{Sym}^2 F} = -2/9$, which are the four values Theorem 12 matches against (18).

B The Weingarten index sums

With $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2 - 1))$, and σ, τ running over S_2 ,

$$\begin{aligned} & \langle U_{i_1 j_1} U_{i_2 j_2} \bar{U}_{i'_1 j'_1} \bar{U}_{i'_2 j'_2} \rangle \\ &= \sum_{\sigma, \tau} \text{Wg}(\sigma \tau^{-1}) \prod_a \delta_{i_a i'_{\sigma(a)}} \delta_{j_a j'_{\tau(a)}}. \end{aligned}$$

Summing the delta products over all N^4 index quadruples,

$$M_{\text{direct}} = \text{Wg}(e)(N^4 + N^2) + 2 \text{Wg}((12))N^3 = N^2,$$

$$M_{\text{cross}} = 2 \text{Wg}(e)N^3 + \text{Wg}((12))(N^4 + N^2) = N.$$

Both are confirmed by explicit summation at $N = 3, 4, 5$.

check: the shared-link weights are Weingarten, not an isotropy assumption

The related fourth moment $\int |U_{11}|^4 dU = 2/[N(N + 1)]$ is $1/6$ at $N = 3$ and in particular nonzero, which is what refuted a claimed Haar vanishing in the balanced $(2, 2)$ sector.

check: $SU(3)$ Weingarten values follow from the general formula

check: the fourth moment integral $|U_{11}|^4 = 1/6$ at $N = 3$

C The $SU(3)$ coefficient ledger

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbour leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

check: the manuscript's $SU(3)$ ledger is this registry, value by value

The u^1 row is $SU(3)$ -only rather than an $SU(3)$ specialisation: it is $-\langle p, -|V|p, - \rangle = 1$, which needs the ϵ channel and vanishes for $N \geq 4$. The other six rows specialise all-rank statements.

D The charge-even ledger

In the same convention, with $\lambda \in \{12, -4\}$ and (19):

Order	Quantity	Value
u^2	hopping $t_{2,+} = \ell_3$	$-11/306$
u^2	on-site diagonal	$223/1020$
u^2	band bottom ($\lambda = 12$)	$-217/1020$
u^2	band top ($\lambda = -4$)	$1109/3060$
u^2	bandwidth $16 t_+ $	$88/153$
u^3	hopping $t_{3,+}$	$-6335/249696$
u^3	band bottom ($\lambda = 12$)	$-54049/520200$
u^3	band top ($\lambda = -4$)	$471353/1560600$

check: one assembly formula gives every C-even value at both orders

Because $t_{r,+} < 0$ at both orders, the $\lambda = -4$ endpoint is the band *maximum* in this sector. A certificate key naming it **bandmin** at both orders is a naming error against its own arithmetic, recorded rather than silently corrected.

check: FINDING: the certificate key 'bandmin' holds the band MAXIMUM, at both orders

E Finite-volume chain calculation

For $i < j$,

$$\partial_2[x; i, j] = [x + e_i; j] - [x; j] - [x + e_j; i] + [x; i],$$

$$\begin{aligned} \partial_3[x; 1, 2, 3] &= [x + e_1; 2, 3] - [x; 2, 3] \\ &\quad - [x + e_2; 1, 3] + [x; 1, 3] \\ &\quad + [x + e_3; 1, 2] - [x; 1, 2]. \end{aligned}$$

Substituting the second into the first cancels every edge pairwise, so $\partial_2 \partial_3 = 0$ over $\mathbb{Z}$.

check: `d_2 d_3 = 0` on the built complex

On the torus $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\text{rank } \partial_3 = L^3 - 1$.

check: `rank d_3 = L^3 - 1` on the built complex

F The two-cube geometry

The eleven oriented plaquettes of the $(3, 2, 2)$ prism, by base vertex and ordered plane axes, are

$$\begin{aligned} & (000; 12), (000; 13), (000; 23), (001; 12), \\ & (010; 13), (100; 12), (100; 13), (100; 23), \\ & (101; 12), (110; 13), (200; 23), \end{aligned}$$

with left, right and shared-face index sets $I_L = (0, 1, 2, 3, 4, 7)$, $I_R = (5, 6, 7, 8, 9, 10)$, $I_F = (7)$. Building $G = BB^\top$ from the oriented boundaries and applying the geometric Möbius fold (16) leaves exactly

four nonzero off-diagonal pairs, $(0, 5), (1, 6), (3, 8), (4, 9)$, each equal to -1 : four disjoint 2×2 blocks and three isolated directions, which is why the spectra in Section 5 are closed form. The $B = 4$ comparator on the same geometry gives $\{0, (-\frac{15}{4})^2, (-\frac{11}{6})^4, (-\frac{5}{3})^4\}$: the paired levels sit *above* the isolated ones, where at $B = 6$ they sit below. That reversal is not the coefficient's sign by itself — within a 2×2 block the pair splits as $D \mp c$, which is invariant under $c \rightarrow -c$ — but the truncation's effect on the connected diagonal as well. What the sign of c alone does reverse is the ordering on a *single* cube, where the kernel is $\alpha I + tG$ with one diagonal and the spectrum of G fixed; that is the shell reversal the one-cube reconstructions report.

check: the B=4 comparator on the same geometry gives
-1/12 with the reversed spectrum

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